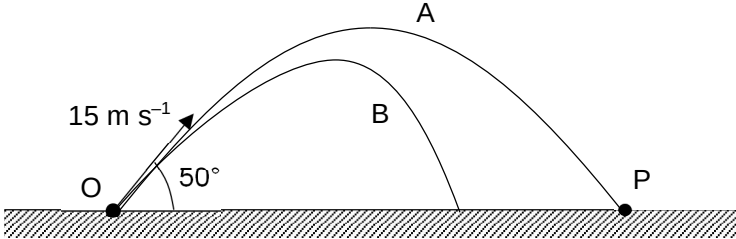


Answers to 2018 JC2 H2 Prelim P3 (H2 Physics)

Suggested Solutions:

No.	Solution	Remarks
1(a)	Vertically, $s_y = u_y t + \frac{1}{2} a_y t^2$ $0 = 15 \sin(50^\circ) t - \frac{1}{2} (9.81) t^2$ Solving, $t = 0$ (not applicable) or 2.34 s. Horizontally, $s_x = u_x t$ $= 15 \cos(50^\circ) (2.34)$ $= 22.6 \text{ m}$ $= 23 \text{ m (2 s.f.)}$ Hence, OP = 23 m	
1(b)(i) 1(b)(ii)	 The diagram illustrates two projectile paths, A and B, starting from point O on a horizontal surface and landing at point P. Path A is the upper trajectory, and path B is the lower trajectory. The initial velocity is labeled as 15 m s^{-1} at an angle of 50° to the horizontal. The horizontal surface is indicated by a hatched line.	
2(a)(i)	$\Delta p = mv - mu$	